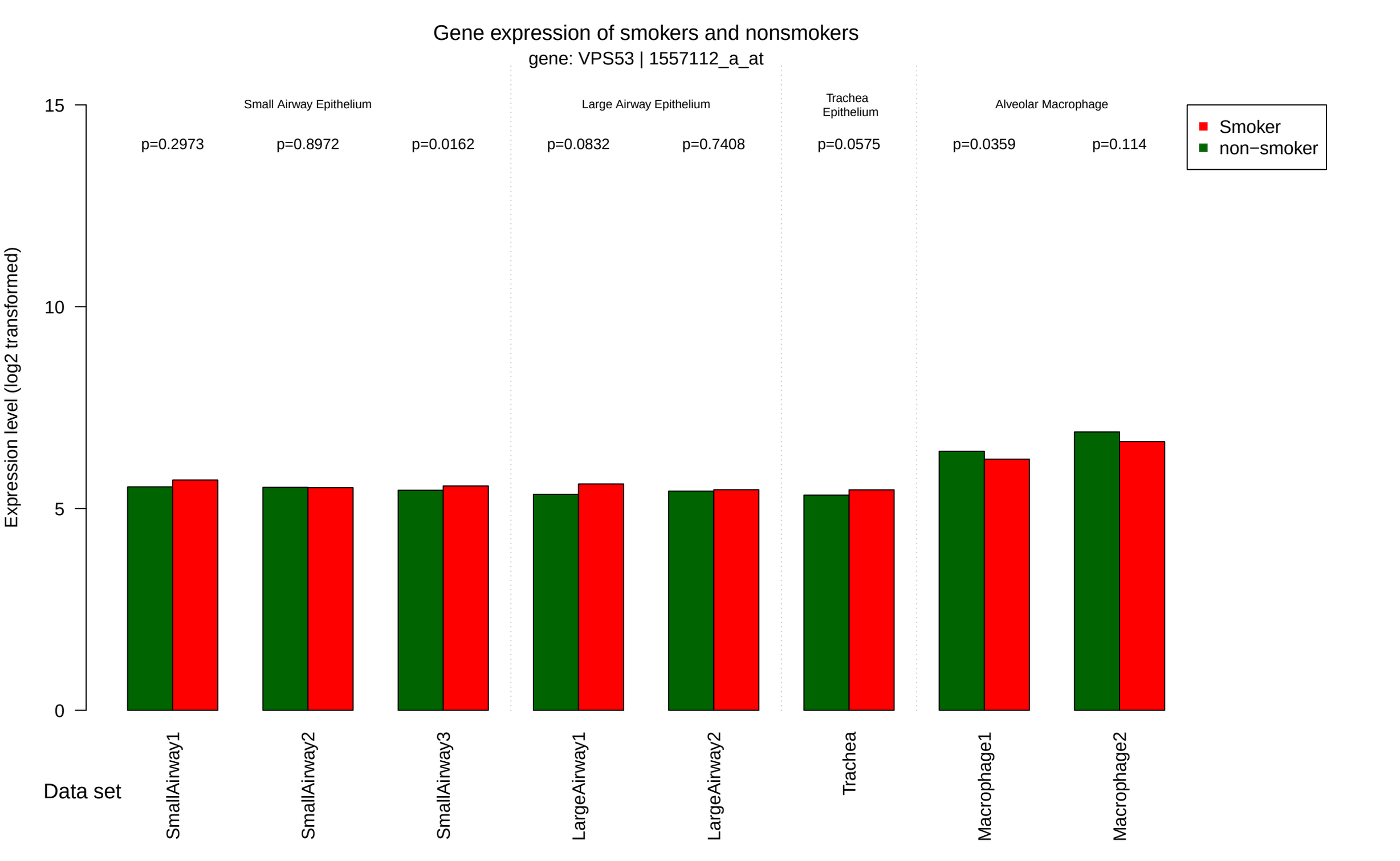
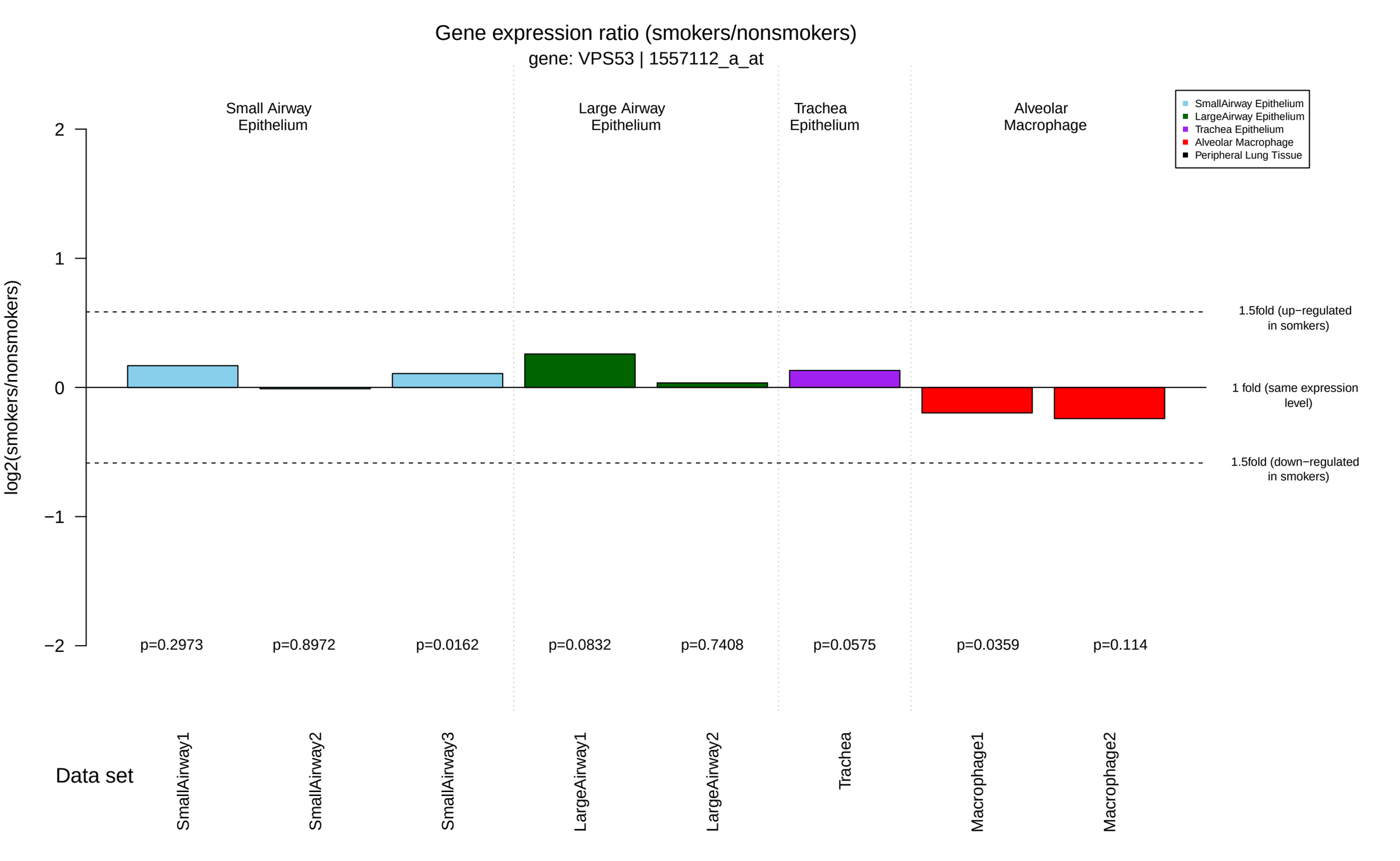
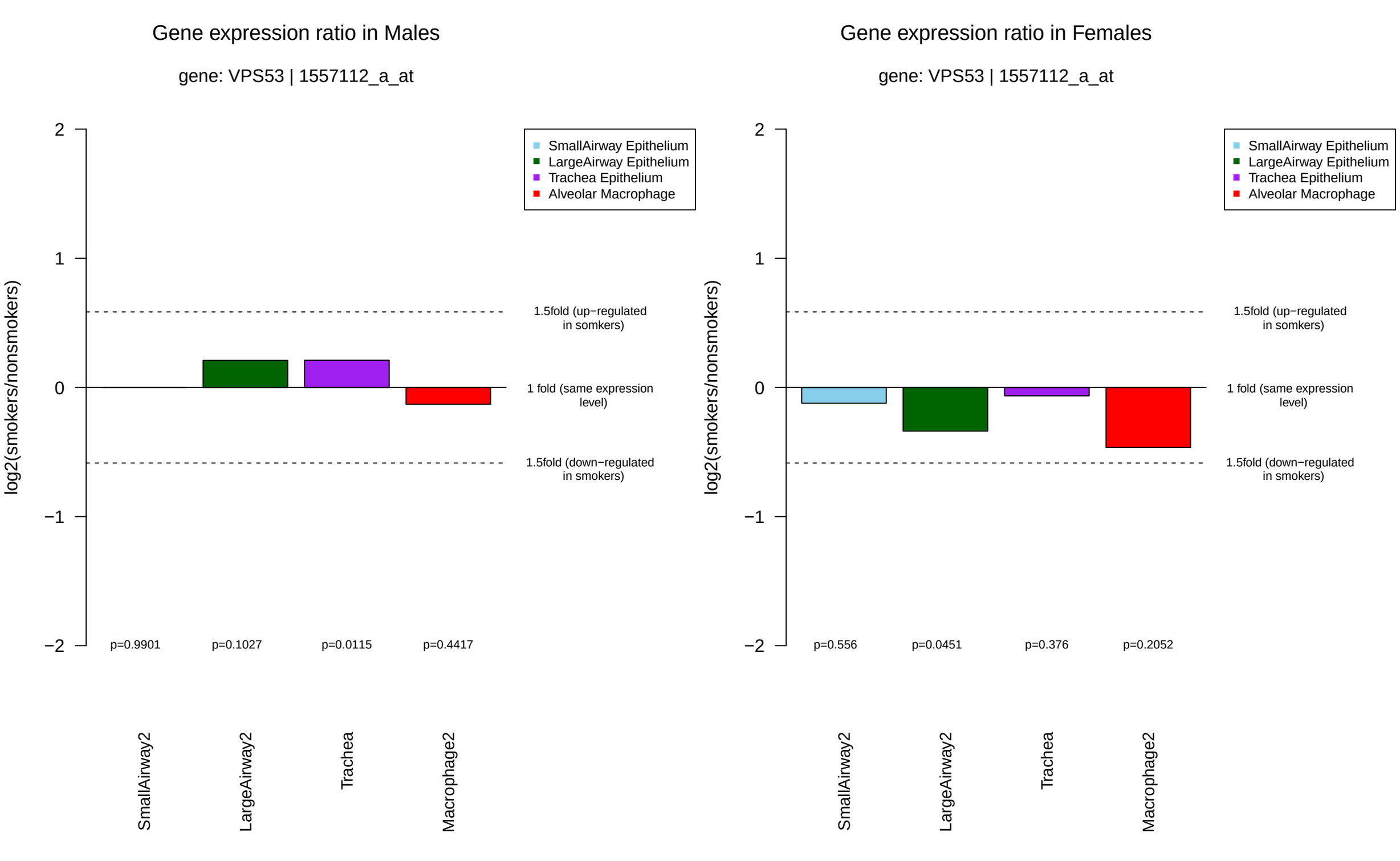
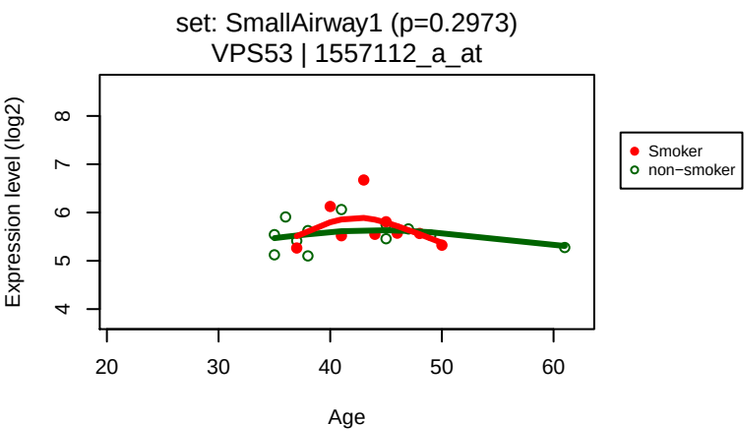
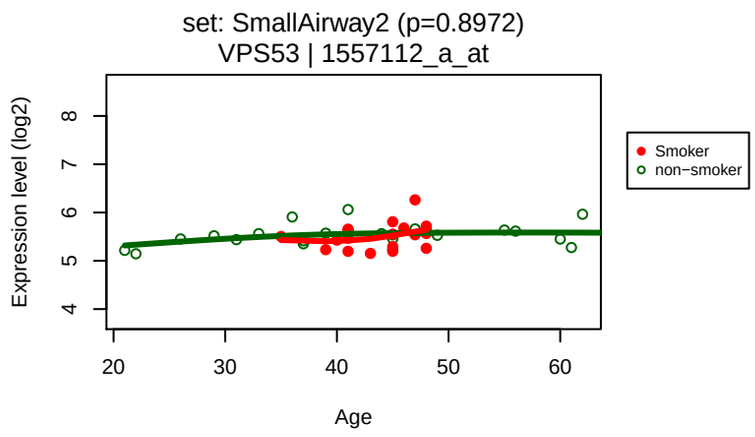
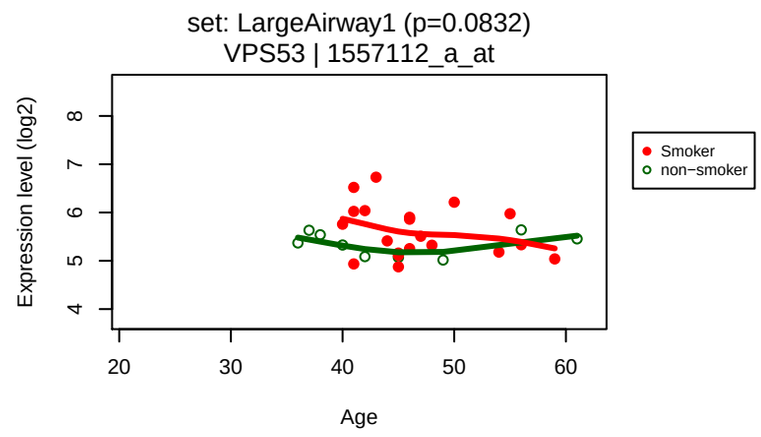

Gene expression of smokers and nonsmokers
gene: VPS53 | 1557112_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2973
p=0.8972
p=0.0162
p=0.0832
p=0.7408
p=0.0575
p=0.0359
p=0.114
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

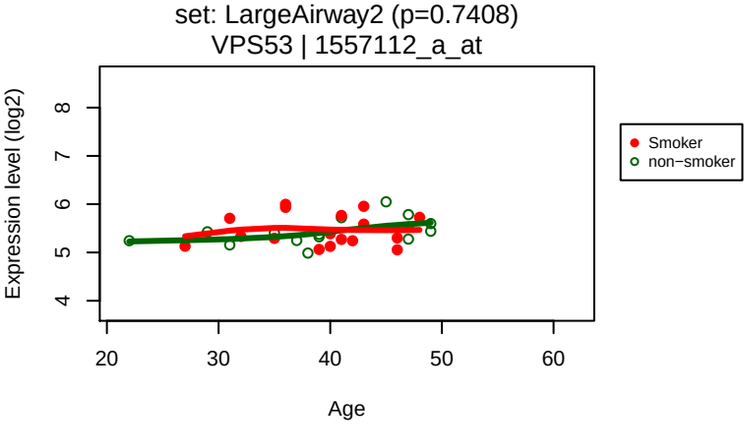

Gene expression ratio (smokers/nonsmokers)
gene: VPS53 | 1557112_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2973
p=0.8972
p=0.0162
p=0.0832
p=0.7408
p=0.0575
p=0.0359
p=0.114
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

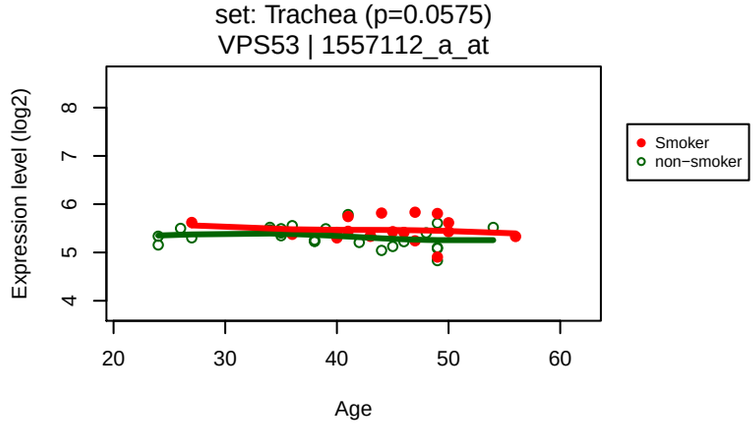

Gene expression ratio in Males
gene: VPS53 | 1557112_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9901
p=0.1027
p=0.0115
p=0.4417
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: VPS53 | 1557112_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.556
p=0.0451
p=0.376
p=0.2052
SmallAirway2
LargeAirway2
Trachea
Macrophage2

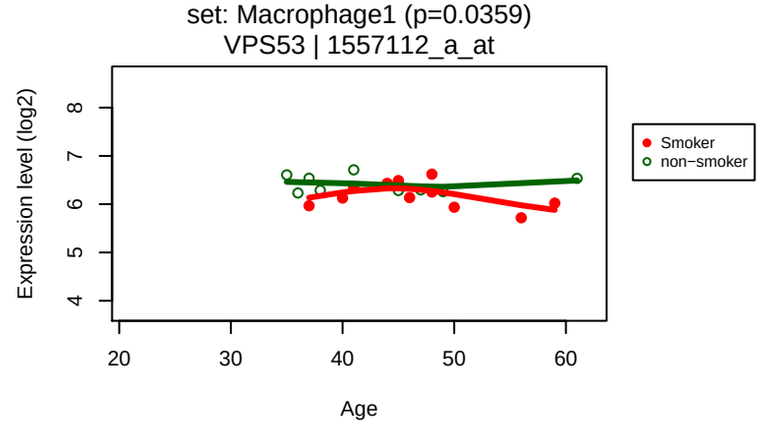

set: SmallAirway1 (p=0.2973)
VPS53 | 1557112_a_at
Expression level (log2)
Age
Smoker
non−smoker

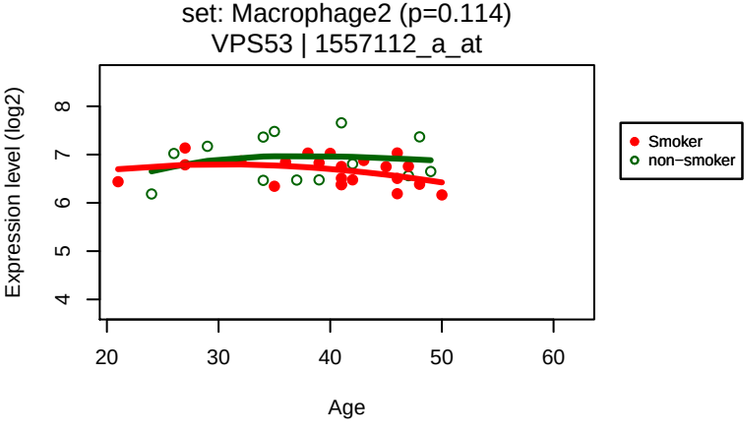

set: SmallAirway2 (p=0.8972)
VPS53 | 1557112_a_at
Expression level (log2)
Age
Smoker
non−smoker

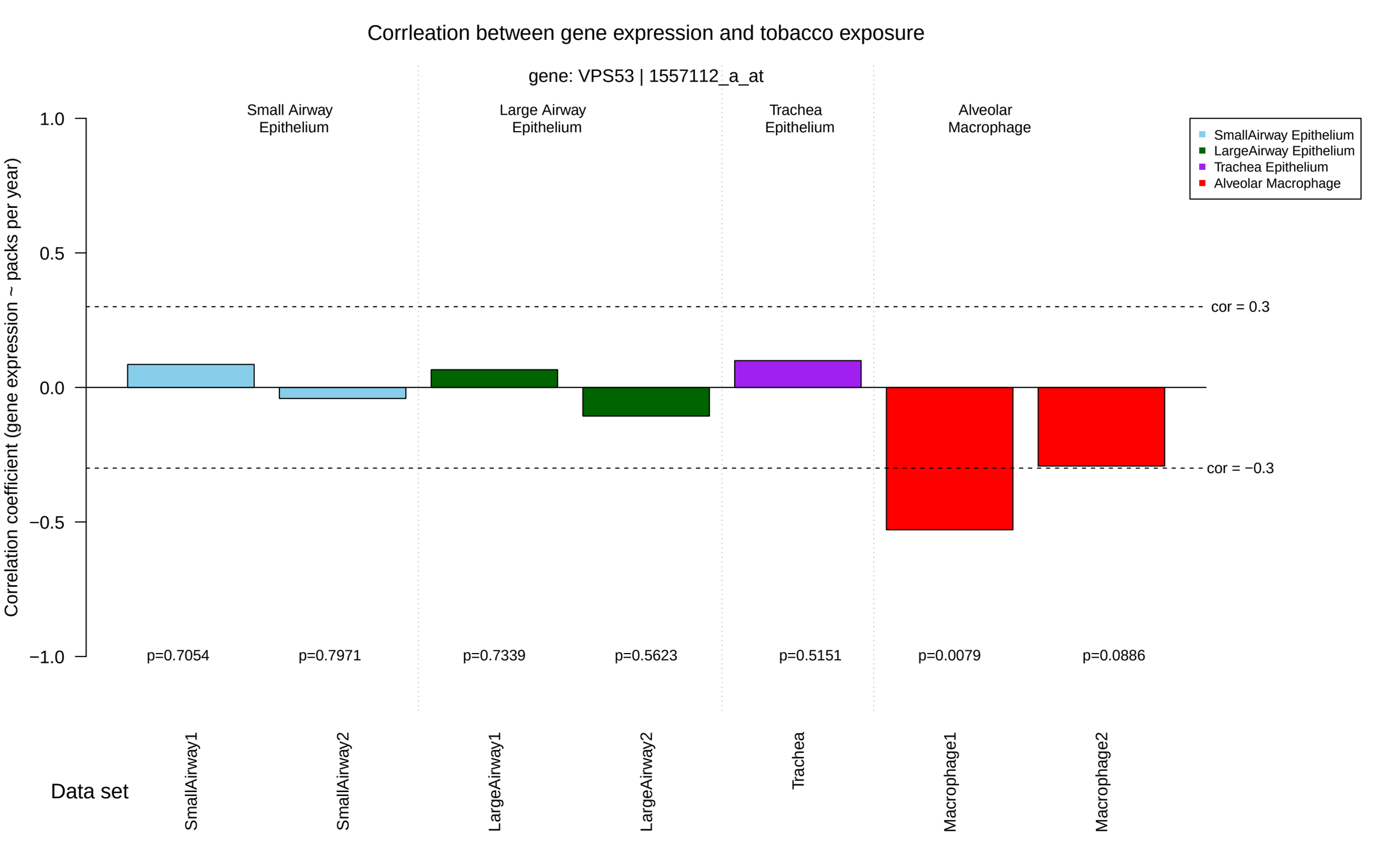

set: LargeAirway1 (p=0.0832)
VPS53 | 1557112_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7408)
VPS53 | 1557112_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0575)
VPS53 | 1557112_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0359)
VPS53 | 1557112_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.114)
VPS53 | 1557112_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: VPS53 | 1557112_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7054
p=0.7971
p=0.7339
p=0.5623
p=0.5151
p=0.0079
p=0.0886
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2